



Linear Workflows

Observe → Act → Update

- One evolving trace
- Simple, fast, lightweight
- Ideal for interactive tasks

1 Loop & Workflow

Plan-Execute Workflows

Plan → Track → Revise

- Make task structure explicit
- Track progress & subgoals
- Replan when needed

Branching Workflows

Sample → Compare → Prune

- Explore multiple candidates
- Choose best / backtrack
- Reduce risk of early mistakes

2 Context & Memory



Working Context

- Discard
- Compress
- Select



Persistent Memory

- Memory Contents
- Memory Operations

3 Tools, MCP and Skills



Tool Interfaces & Protocols



Active Tool Discovery



Skill Libraries

4 Orchestration



Decomposition and Roles



Coordination Topologies



Orchestration Optimization



Agent Protocols

5 Hooks & Middleware



Pre-defined Ruled-based

- Input boundary
- Tool-call boundary
- Protocol boundary



Custom User-defined

- Symbolic
- Static LLM-compiled
- Dynamic LLM-evaluated



Runtime-adaptive

- Uncertainty-driven
- Behavior-driven
- Threat-shift-driven

6 Verification

Verification Levels



- step-level
- end-of-run level



Verifier Strategies

- Auto labels & rewards
- Verifier-guided search
- Compute-optimal verification



Assessment Targets

- correctness
- safety
- faithfulness
- multi-agent



Set Goal



Understand & Plan



Execute & Interact



Reflect & Learn



Adapt & Recover



Complete & Deliver